



American International University-Bangladesh (AIUB)

Department of Computer Science

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AR Based Clothing Try-On System

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1. PROJECT PROPOSAL

1.1 Background to the Problem

The rapid growth of online fashion retail has made e-commerce dominant, yet it faces 20–40% return rates primarily due to customers' inability to try on garments before purchase, leading to fit, style, and appearance dissatisfaction. Traditional tools like size charts, static images, and reviews fall short against brand-specific sizing variations, diverse body types, and subjective style preferences, forcing guesswork, multi-size orders, and high returns.

This impacts stakeholders severely: retailers face reverse logistics, restocking, and inventory depreciation costs; customers endure frustration and lost trust; and excessive returns generate environmental waste from shipping/packaging. The AR-Based Clothing Try-On System addresses this by enabling real-time virtual try-ons via smartphone cameras, reducing returns by 25–30%, boosting conversions, and promoting sustainable practices through better purchase confidence.

1.2 Selection of Process Model

Scrum is selected for the AR-Based Clothing Try-On System because it best matches the project's high uncertainty in computer vision/AR development, iterative feature delivery needs (per PRD phases), and need for continuous retailer/customer feedback.

Key Scrum Roles:

- Product Owner: Prioritizes PRD features (Customer/Retailer stories)
- Scrum Master: Removes blockers (AR integration, 3D model issues)
- Development Team: Delivers working AR try-on increments

Key Artifacts:

- Product Backlog: All PRD user stories from previous table
- Sprint Backlog: Stories for current sprint (e.g., "AR Camera + Body Detection")
- Increment: Working AR try-on feature at sprint end

Model	Why NOT Selected	Why Scrum is Better
Waterfall	Requirements freeze too early; AR/computer vision needs iteration	Iterative sprints allow body tracking refinement based on real device testing
V-Model	Sequential testing doesn't fit rapid AR prototype → feedback cycles	Continuous integration/testing per sprint
Prototyping	Good for UI but misses structured delivery of PRD phases	Delivers complete increments (Phase 1 MVP) every 2 weeks
XP (Extreme Programming)	Too developer-focused; needs retailer stakeholder involvement	Scrum ceremonies include business demos
DSDM	Complex enterprise focus; this is startup-scale project	Simpler Scrum better for small team
FDD	Feature-driven but ignores cross-cutting concerns (auth, analytics)	Scrum handles user stories across all layers

Specific Reasons for Scrum:

1. **Uncertainty Management:** AR body tracking accuracy unknown → iterative refinement
2. **Phased PRD Alignment:** Sprint 1 = Core AR Try-On; Sprint 2 = Size Recs; Sprint 3 = Retailer Dashboard
3. **Stakeholder Feedback:** Retailers test 3D models, Customers validate UX every sprint
4. **Small Team Fit:** 4-6 developers can deliver 30 FPS AR increments bi-weekly
5. **Risk Reduction:** Early detection of computer vision bottlenecks vs. Waterfall's late discovery

Expected Timeline: 6 sprints (12 weeks) for Phase 1 MVP delivery.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 1		Auth Feature				
phase 1	1	Sign Up [Customer] [Retailer] [Admin]	<pre> <SignUpForm> <input=name> <input=email> <input=phone> <input=gender> <input=date-of-birth> <input=password> <input=confirm-password> <dropdown=user-type> <!-- Customer/Retailer/Admin --> <checkbox=terms-and-conditions> <button=sign-up> <link=sign-in> <link=privacy-policy> </SignUpForm> </pre>	As a new user, I want to create an account with my email or phone number so that I can access the AR try-on platform securely and save my preferences.	The email or phone number must be unique and valid; if already registered, the system should display "Email/phone already in use" error message. Password must be at least 8 characters long, and if this requirement is not met, an invalid password message should be shown. The confirm password field must match the password field exactly, otherwise an error should be displayed. User type selection is mandatory and must be chosen from the available options (Customer, Retailer, or Admin). Users must accept the terms and conditions before proceeding. Upon successful registration, the user should be redirected to their respective dashboard based on user type, and a confirmation email or SMS should be sent to verify their account.	signup
phase 1	2	Sign In [Customer] [Retailer] [Admin]	<pre> <SignInForm> <input=email-or-phone> <input=password> <checkbox=remember-me> <button=sign-in> <link=forget-password> <link=sign-up> <divider> <button=sign-in-with-google> <button=sign-in-with-facebook> </SignInForm> </pre>	As a registered user, I want to sign in to my account using my email/phone and password so that I can access my saved preferences, measurements, and shopping history.	The system must validate that the entered email or phone number exists in the database; if not found, display "Account not found" error. The password must match the stored password for the given account; if incorrect, show "Invalid password" error after a maximum of 3 failed attempts, temporarily lock the account for security. If "Remember me" is checked, the user session should persist for 30 days. Upon successful login, redirect users to their role-specific dashboard (Customer to shopping interface, Retailer to product management, Admin to system controls). The system should support social login options (Google, Facebook) as alternative sign-in methods.	signin
phase 1	3	Forgot Password [Customer] [Retailer] [Admin]	<pre> <ForgotPasswordForm> <input=email-or-phone> <button=send-reset-link> <link=back-to-sign-in> <text=succes-message> </ForgotPasswordForm> <ResetPasswordForm> <input=new-password> <input=confirm-new-password> <button=reset-password> </ResetPasswordForm> </pre>	As a user who forgot my password, I want to receive a password reset link via email or SMS so that I can regain access to my account securely.	The system must verify that the entered email or phone number is registered; if not found, display "No account found with this email/phone" message. A password reset link or OTP should be sent within 1 minute to the registered email or phone number. The reset link should expire after 15 minutes for security purposes. The new password must meet minimum security requirements (at least 8 characters, one uppercase, one number, one special character). The confirm password field must match the new password exactly. Upon successful password reset, the user should be redirected to the sign-in page with a "Password successfully reset" confirmation message, and all active sessions should be invalidated for security.	recovery
phase 1	4	Role-Based Access Control [Admin]	<pre> <AccessControlView> <section=role-permissions> <text=current-role> <list=permissions> </section> <section=restricted-access-message> <icon=access-denied> <text=error-message> <button=back-to-dashboard> </section> </AccessControlView> </pre>	As a system administrator, I want different user types to have appropriate access levels so that customers, retailers, and admins can only access features relevant to their roles.	Customer users should have access to browsing products, virtual try-on, shopping cart, order management, and profile settings, but not to product upload or system configuration. Retailer users should have access to product catalog management, 3D model uploads, inventory management, sales analytics, and customer feedback, but not to system-wide configuration or user management. Admin users should have full access to all features including user management, system configuration, platform analytics, content moderation, and retailer approval. When a user attempts to access a restricted feature, the system must display "Access Denied: You do not have permission to view this page" and redirect them to their appropriate dashboard. All role assignments	verify

Feature: 2		AR Try-On Core Functionality				
phase 1	5	Real-time AR Garment Overlay on Live Camera Feed [Customer]	<pre> <CameraView> <camera-feed /> <ar-overlay /> <garment-layer /> <control-panel> <button=capture> <button=switch-camera> <button=toggle-flash> </control-panel> <product-info-card> <text=product-name> <text=price> <button=add-to-cart> </product-info-card> </CameraView> </pre>	As a shopper, I want to see clothing items overlaid on my live camera feed in real-time so that I can visualize how they look on me before purchasing.	The camera feed must display immediately upon feature activation, with the AR garment overlay appearing within 2 seconds of product selection. The overlay should update at a minimum of 30 fps to ensure a smooth and responsive experience, and the garment must remain visible and properly aligned throughout the entire session. If camera permission is denied by the user, the system should display an error message with clear instructions guiding them to the device settings to enable camera access.	Menu
phase 1	6	Automatic Body Detection and Tracking [Customer]	<pre> <BodyDetectionView> <camera-feed /> <body-skeleton-overlay /> <detecton-status-indicator> <icon=body-detected> <text=status-message> </detecton-status-indicator> <positioning-guide> <text=instruction> <visual-frame /> </positioning-guide> <button=retry-detection> </BodyDetectionView> </pre>	As a user, I want the system to automatically detect and track my body position so that I don't have to manually adjust or calibrate the AR experience.	Body detection must complete within 3 seconds of camera activation and successfully identify key body points including shoulders, waist, hips, and neck. The system should maintain detection accuracy across various lighting conditions with a minimum accuracy rate of 90% for standard poses. If the body cannot be detected, the system must display a positioning guide with clear instructions to help the user adjust their position. Additionally, contextual messages such as "Move closer" or "Step back" should appear when the user is out of the optimal detection range.	dashboard
phase 1	7	Realistic 3D Garment Rendering [Customer]	<pre> <RenderingView> <camera-feed /> <3d-garment-model /> <rendering-settings> <slider=lighting-intensity> <toggle=shadows> <toggle=reflections> <dropdown=quality-preset> </rendering-settings> <material-preview> <text=fabric-type> <visual=texture-sample> </material-preview> <button=apply-settings> </RenderingView> </pre>	As a shopper, I want to see realistic 3D rendering of clothing with accurate colors, textures, and fabric behavior so that I can make confident purchase decisions based on true-to-life visualization.	Garments must display with high-resolution textures of at least 1024x1024 pixels, with colors matching the actual product within 95% accuracy. Fabric properties such as silk drape or denim stiffness should be rendered with realistic physics simulations. The lighting system must adapt to the user's environment to create natural-looking garment appearance, and shadows and wrinkles should appear appropriately based on body position and garment type. The rendering quality should adjust automatically based on device performance capabilities while maintaining a minimum acceptable quality level even on mid-range smartphones to ensure consistent user experience across different devices.	credits

Feature: 3		Body Measurement & Fit Analysis				
phase 1	8	Automated Body Measurement Detection [Customer]	<pre> <BodyMeasurementView> <camera-feed /> <measurement-overlay /> <instruction-panel> <text=step-by-step-guide> <icon=pose-illustration> </instruction-panel> <measurement-display> <text=height> <text=chest> <text=waist> <text=hips> <text=shoulder-width> </measurement-display> <button=save-measurements> <button=retake-measurements> </BodyMeasurementView> </pre>	As a customer, I want the system to automatically detect my body measurements using the camera so that I don't have to manually measure and input my dimensions.	The system must capture and calculate body measurements within 10 seconds of completing the guided poses. Key measurements including height, chest, waist, hips, and shoulder width must be detected with at least 95% accuracy compared to manual measurements. Clear step-by-step instructions should guide users through proper positioning and posture for accurate measurement capture. If measurements fall outside realistic human ranges, the system should prompt the user to retake measurements. All detected measurements should be automatically saved to the user's profile and remain editable for manual corrections if needed.	complaints
phase 1	9	Size Recommendation Based on Measurements [Customer]	<pre> <SizeRecommendationView> <product-image /> <size-chart /> <recommended-size-card> <text=recommended-size> <text=confidence-level> <text=reason> </recommended-size-card> <alternative-sizes> <card=size-option-1> <card=size-option-2> </alternative-sizes> <button=select-size> <link=view-full-size-chart> </SizeRecommendationView> </pre>	As a shopper, I want to receive personalized size recommendations based on my body measurements so that I can order the correct size without guesswork.	The system must provide a size recommendation within 1 second of viewing a product, using the user's saved body measurements and the retailer's size chart. The recommendation should include a confidence level percentage (e.g., "90% confident this will fit") and a brief explanation of why this size was selected. Alternative size suggestions should be displayed if the user's measurements fall between standard sizes or if fit preferences vary. The system should account for different sizing standards across brands and regions. If body measurements are not available, the system should prompt the user to complete the measurement process before providing recommendations.	tasks
phase 1	10	Measurement Accuracy Calibration [Customer]	<pre> <CalibrationView> <calibration-instructions> <text=place-reference-object> <image=credit-card-example> </calibration-instructions> <camera-feed /> <reference-object-detector /> <calibration-status> <text=calibration-progress> <progress-bar /> </calibration-status> <calibration-results> <text=accuracy-score> <text=calibration-quality> </calibration-results> <button=complete-calibration> <button=skip-calibration> </CalibrationView> </pre>	As a user, I want to calibrate the measurement system using a reference object so that my body measurements are as accurate as possible.	The system must guide users to place a standard reference object (such as a credit card) in the camera view for scale calibration before taking body measurements. Calibration should complete within 5 seconds of detecting the reference object and provide feedback on calibration quality. The system must adjust all subsequent measurements based on the calibration data to ensure accuracy across different camera setups and distances. If calibration fails or produces low-quality results, clear instructions should be provided to help users retry with better positioning or lighting. Users should have the option to skip calibration, but the system must display a warning that measurement accuracy may be reduced without proper calibration.	

Feature: 4		Product Visualization Controls				
phase 1	11	Multiple Viewing Angles (360-degree) [Customer]	<pre> <ViewingAngleControl> <camera-feed /> <ar-garment-layer /> <rotation-controls> <button=rotate-left> <button=rotate-right> <slider=rotation-angle> </rotation-controls> <view-presets> <button=front-view> <button=side-view> <button=back-view> </view-presets> <angle-indicator> <text=current-angle> <icon=compass> </angle-indicator> </ViewingAngleControl> </pre>	As a customer, I want to view the virtual garment from multiple angles including front, side, and back views so that I can see how it looks from all perspectives.	The system must allow users to rotate their view 360 degrees around the garment either by swiping on the screen or using rotation buttons. The rotation should be smooth with no lag, updating at minimum 30 fps. Quick access buttons for standard views (front, side, back) should instantly switch to those angles within 0.5 seconds. The current viewing angle should be displayed as a degree measurement or compass indicator. Users should be able to manually rotate the view using touch gestures with responsive feedback, and the garment must remain properly aligned with the body during all rotations.	payments
phase 1	12	Zoom In/Out Functionality [Customer]	<pre> <ZoomControl> <camera-feed /> <ar-garment-layer /> <zoom-controls> <button=zoom-in> <button=zoom-out> <button=reset-zoom> </zoom-controls> <zoom-level-indicator> <text=zoom-percentage> <progress-bar /> </zoom-level-indicator> <pinch-gesture-hint /> </ZoomControl> </pre>	As a user, I want to zoom in on specific parts of the garment to examine details like fabric texture, stitching, and patterns more closely.	The system must support pinch-to-zoom gestures with smooth scaling from 100% to 300% magnification without pixelation or quality loss. Zoom controls should include both gesture-based input and on-screen buttons for accessibility. The current zoom level should be displayed as a percentage with a visual indicator. When zoomed in, users should be able to pan across the garment by dragging to view different areas in detail. A reset button must be available to instantly return to the default 100% zoom level. The zoom functionality should maintain AR tracking accuracy even at maximum magnification.	licensing
Feature: 5		Try-On Session Management				
phase 1	13	Save Current Try-On Session [Customer]	<pre> <SessionSaveView> <current-try-on-preview /> <session-details> <text=product-name> <text=selected-size> <text=selected-color> <text=date-time> </session-details> <save-options> <input=session-name> <button=save-to-favorites> <button=add-to-wishlist> </save-options> <button=save-session> <button=discard-session> </SessionSaveView> </pre>	As a customer, I want to save my current try-on session including the product, size, and color selections so that I can return to it later without starting over.	The system must allow users to save their current try-on session at any time with a single button click. Saved sessions should include all relevant information including product details, selected size, color variant, any customization settings, and a timestamp. Users should be able to provide an optional custom name for their saved session for easy identification. The save functionality should complete within 2 seconds and display a confirmation message. All saved sessions must be accessible from the user's profile or a dedicated "Saved Try-Ons" section. Sessions should remain saved until manually deleted by the user or until a maximum storage limit is reached with appropriate notification.	credits
phase 1	14	Capture Screenshots [Customer]	<pre> <ScreenshotView> <camera-feed /> <ar-garment-layer /> <capture-button> <icon=camera> </capture-button> <screenshot-preview> <thumbnail=captured-image> </screenshot-preview> <screenshot-actions> <button=save-to-device> <button=share> <button=delete> </screenshot-actions> <gallery-link> <text=view-all-screenshots> </gallery-link> </ScreenshotView> </pre>	As a user, I want to capture screenshots of myself wearing virtual garments so that I can save them for later reference or share them with friends for opinions.	The system must provide a clearly visible capture button that takes a high-resolution screenshot of the current AR view within 0.5 seconds of being pressed. Screenshots should be saved in a standard image format (JPEG or PNG) with resolution matching the device's camera quality. A thumbnail preview of the captured screenshot should appear immediately after capture with options to save, share, or delete. All screenshots should be automatically stored in a gallery within the app that users can access at any time. The system should handle storage permissions appropriately and display clear error messages if storage is unavailable or full. Captured screenshots should include metadata such as product name, date, and time for organization purposes.	redemption
	15	Record Try-On Videos [Customer]	<pre> <VideoRecordingView> <camera-feed /> <ar-garment-layer /> <recording-controls> <button=start-recording> <button=stop-recording> <button=pause-recording> </recording-controls> <recording-indicator> <icon=recording-status> <text=recording-duration> </recording-indicator> <video-preview> <thumbnail=recorded-video> </video-preview> <video-actions> <button=save-video> <button=share-video> <button=delete-video> </video-actions> </VideoRecordingView> </pre>	As a customer, I want to record videos of myself wearing virtual clothing while moving so that I can see how the garment looks and moves from different angles in motion.	The system must provide video recording functionality with clear start, stop, and pause controls. Recording should support a minimum duration of 30 seconds and maximum of 60 seconds with a visible timer displaying elapsed time. The recorded video should maintain a minimum frame rate of 30 fps with synchronized AR garment overlay throughout. A visual recording indicator (red dot or similar) must be clearly visible during active recording. After recording stops, an immediate preview of the video should be available with options to save to device, share on social media, or delete. Videos should be saved in a standard format (MP4 or similar) compatible with most devices and social platforms. The system should check available storage before recording and warn users if insufficient space is available.	
	16	Try-On History Tracking [Customer]	<pre> <TryOnHistoryView> <history-list> <card=try-on-session-1> <thumbnail /> <text=product-name> <text=date-time> <button=view-details> </card> <!-- More history cards --> </history-list> <filter-options> <dropdown=sort-by> <!-- Recent, Favorites, Product Type --> <dropdown=date-range> </filter-options> <search-bar> <input=search-products> </search-bar> <button=clear-history> </TryOnHistoryView> </pre>	As a user, I want to view my complete try-on history so that I can revisit previously tried products and make purchase decisions based on my past exploration.	The system must automatically track and store all try-on sessions with timestamps, product details, and thumbnail images. The history view should display sessions in reverse chronological order by default with most recent tries appearing first. Users should be able to filter history by date range, product category, or favorites status. A search function must allow users to quickly find specific products by name or keywords. Each history entry should be clickable to view full details and instantly reload that specific try-on session with all previous settings restored. The system should maintain try-on history for at least 90 days with options to manually delete individual entries or clear entire history. If a product from history is no longer available, it should be clearly marked as "Out of Stock" or "Unavailable" but remain viewable in the history for reference purposes.	

Feature: 6		Product Catalog & Selection				
phase 1	17	Browse Available Clothing Items [Customer] [Retailer]	<pre> <ProductCatalogView> <navigation-bar> <dropdown-category> <!-- Tops, Bottoms, Dresses, etc --> <input-search> </navigation-bar> <product-grid> <card-product-1> <image-product-image> <text-product-name> <text-price> <badge-new-arrival> <button-quick-view> <button-try-on-ar> </card> <!-- More product cards --> </product-grid> <pagination> <button-previous-page> <text-page-number> <button-next-page> </pagination> </ProductCatalogView> </pre>	As a customer, I want to browse through all available clothing items in an organized catalog so that I can discover products I'm interested in trying on virtually.	The product catalog must display all available items in a responsive grid layout with clear product images, names, and prices visible for each item. Products should be organized by categories (tops, bottoms, dresses, outerwear, etc.) with easy navigation between categories through a dropdown or tab menu. Each product card should include a "Try On AR" button for immediate virtual try-on access and a "Quick View" option for seeing more details without leaving the catalog page. The catalog should load quickly with pagination or infinite scroll to handle large inventories efficiently, loading no more than 20-30 products per page. New arrivals, sale items, and trending products should be clearly marked with badges or labels. The interface must be responsive and work seamlessly on both mobile and desktop devices.	ledger
phase 1	18	Product Filtering and Search [Customer] [Retailer]	<pre> <FilterSearchView> <search-bar> <input-search-query> <button-search> <icon-voice-search> </search-bar> <filter-panel> <section-category-filter> <checkbox>tops> <checkbox>bottoms> <checkbox>dresses> </section> <section-price-range> <slider-min-price> <slider-max-price> </section> <section-size-filter> <checkbox>XS> <checkbox>S> <checkbox>M> <checkbox>L> <checkbox>XL> </section> <section-color-filter> <checkbox>black> <checkbox>white> <checkbox>blue> <!-- More colors --> </section> </filter-panel> </FilterSearchView> </pre>	As a customer, I want to filter and search products by various criteria so that I can quickly find clothing items that match my preferences and needs.	The system must provide a comprehensive search function that accepts text queries and returns relevant results within 2 seconds based on product names, descriptions, categories, and tags. Multiple filter options should be available including category, price range, size, color, brand, style, and material with the ability to apply multiple filters simultaneously. Active filters should be clearly displayed as removable chips or badges above the product results. Filter results should update immediately as users make selections without requiring a separate "apply" action unless dealing with complex filters. A "Clear All" button must be available to reset all filters at once. The system should display the number of results matching current filters and show a "No results found" message with suggestions if no products match. Search history and popular searches should be suggested when users click the search bar.	analytics
	19	Quick Add to Try-On Queue [Customer]	<pre> <TryOnQueueView> <queue-header> <text-items-in-queue> <button-clear-queue> </queue-header> <queue-list> <card-queued-item-1> <thumbnail /> <text-product-name> <text-size> <text-color> <button-remove> </card> <!-- More queued items --> </queue-list> <quick-add-section> <text-suggested-products> <card-suggestion-1> <button-add-to-queue> </card> <!-- quick-add-section --> </quick-add-section> <button-start-try-on-session> </TryOnQueueView> </pre>	As a shopper, I want to quickly add multiple products to a try-on queue so that I can efficiently try them on one after another without searching for each item individually.	The system must allow users to add products to a try-on queue from any page in the catalog with a single button click. The queue should display all added items with thumbnails, names, and selected options (size, color) in a sidebar or dedicated view. Users should be able to reorder items in the queue by dragging or using move up/down buttons to prioritize their try-on sequence. Each queued item should have a remove button for easy deletion from the queue. The queue should persist across sessions until items are manually removed or tried on. A counter badge should display the number of items in the queue and be visible from any page. When users start a try-on session, the system should load items from the queue in order, allowing seamless switching between queued products. The queue should support a maximum of 20 items with a notification when the limit is reached.	
Feature: 7		User Interface & Experience				
phase 1	20	Intuitive Camera Controls [Customer]	<pre> <CameraControlView> <camera-feed /> <camera-controls> <button-switch-camera> <button-flash-toggle> <slider-exposure> </camera-controls> <gesture-hints> <text-pinch-to-zoom> <text-swipe-to-rotate> </gesture-hints> <tutorial-overlay> <text-instructions> <button-got-it> </tutorial-overlay> </CameraControlView> </pre>	As a user, I want easy-to-use camera controls so that I can adjust settings without confusion during my AR try-on experience.	Camera controls must be intuitive with universally recognizable icons positioned in easily reachable areas for one-handed operation on mobile devices. First-time users should see a brief tutorial overlay explaining gesture controls that can be dismissed and accessed again from settings. The switch camera button should toggle between front and rear camera within 1 second with smooth transition, and flash toggle should clearly indicate on/off status. All camera controls should have sufficient size and spacing to prevent accidental taps, and control visibility should auto-fade when inactive but reappear with any user interaction.	compliance
phase 1	21	Simple Navigation Menu [Customer] [Retailer] [Admin]	<pre> <NavigationMenu> <header> <logo /> <button-menu-toggle> <icon-shopping-cart> </header> <sidebar-menu> <item-home> <item-browse-products> <item-ar-try-on> <item-wishlist> <item-profile> <item-help> </sidebar-menu> <role-specific-items> <!-- Retailer/Admin specific items --> </role-specific-items> </NavigationMenu> </pre>	As a user, I want a simple and consistent navigation menu so that I can easily access different features and sections of the application without getting lost.	The navigation menu must be accessible from every page through a clearly visible menu icon with items organized logically using clear labels and icons. The current active page should be visually highlighted, and role-specific menu items should be displayed based on user type with appropriate access controls. On mobile devices, the menu should use a hamburger icon or bottom navigation with no more than 5 primary items for optimal usability, and the menu should close automatically when a user selects an item or taps outside the menu area.	legal
phase 1	22	Help and Tutorial Guides [Customer] [Retailer] [Admin]	<pre> <HelpView> <search-help> <input-search-topics> </search-help> <help-categories> <card-getting-started> <card-ar-guide> <card-troubleshooting> </help-categories> <faq-section> <accordion-faq-items> </faq-section> <button-contact-support> </HelpView> </pre>	As a user, I want access to help guides and tutorials so that I can learn how to use the AR try-on features effectively and troubleshoot any issues I encounter.	The help section must be easily accessible from the main navigation menu and include a searchable knowledge base with articles covering all major features. Interactive tutorials should be available for first-time users walking them through the AR try-on process with visual demonstrations, and video tutorials should demonstrate key features with videos no longer than 2 minutes each. A comprehensive FAQ section should address common issues organized by category, and contact options for live support should be prominently displayed with estimated response times clearly stated.	ops
phase 1	23	Error Handling and Feedback Messages [Customer] [Retailer] [Admin]	<pre> <ErrorHandlerView> <error-modal> <icon-error-type> <text-error-message> <text-suggested-action> <button-retry> </error-modal> <success-notification> <icon-success> <text-message> </success-notification> <loading-indicator> <spinner /> <text-loading-message> </loading-indicator> </ErrorHandlerView> </pre>	As a user, I want clear and helpful error messages when something goes wrong so that I understand what happened and know how to fix it or proceed.	Error messages must be written in plain, user-friendly language and clearly explain what went wrong with specific suggested actions the user can take to resolve the issue. Different error types should be visually distinguished using color coding and icons, and success messages should appear briefly after successful actions. Loading indicators should be displayed during any operation taking longer than 1 second, and network connectivity issues should trigger a persistent banner message. Critical errors should provide options to retry the operation, contact support, and all error messages should be logged for troubleshooting purposes while maintaining user privacy.	performance

Feature: 8		Social & Sharing Features				
phase 2	24	Share Try-On Results on Social Media [Customer]	<SocialShareView> <try-on-preview /> <share-options> <button=share-instagram> <button=share-facebook> <button=share-tiktok> <button=share-whatsapp> <button=share-twitter> </share-options> <caption-input> <textarea=custom-caption> </caption-input> <link-generation> <text=shareable-link> <button=copy-link> </link-generation> </SocialShareView>	As a customer, I want to share my virtual try-on results on social media.	Share screenshots/videos to Instagram, Facebook, TikTok, WhatsApp, Twitter directly. Auto-generate captions with product info and affiliate links. Create shareable links. Track shares for analytics. Pre-fill with product hashtags.	social
phase 2	25	Friend Opinion and Feedback System [Customer]	<FriendFeedbackView> <try-on-image /> <feedback-request> <input=friend-phone-or-email> <text=custom-message> <button=send-request> </feedback-request> <feedback-received> <card=friend-1> <text=friend-name> <rating=thumbs-up-down> <text=comment> <text=timestamp> </card> </feedback-received> <feedback-summary> <chart=sentiment-analysis> </feedback-summary> </FriendFeedbackView>	As a customer, I want friends' opinions on my virtual try-on choices.	Send feedback requests via SMS/email. Friends rate (thumbs up/down) and comment. Receive notifications of responses. Display feedback summary with sentiment analysis. Preserve anonymity option. Remind non-respondents after 24 hours.	social
Feature: 9		Advanced Customization & Personalization				
phase 2	26	Garment Color and Pattern Variants [Customer] [Retailer]	<ColorPatternView> <garment-preview /> <color-palette> <color-switch=color-1> <color-switch=color-2> <color-switch=color-3> </color-palette> <pattern-options> <button=solid> <button=stripes> <button=checks> <button=floral> <button=custom-pattern> </pattern-options> <real-time-preview> <text=selected-color> <text=selected-pattern> </real-time-preview> <button=save-variant> </ColorPatternView>	As a shopper, I want to see how different colors and patterns look on me.	Display all available color variants in real-time. Support pattern visualization (stripes, checks, florals). Real-time preview update within 500ms. Save favorite combinations. Show color accuracy within 95%. Display price differences for variants.	customization
phase 2	27	Fabric Material Simulation [Customer] [Designer]	<MaterialSimView> <3d-garment-model /> <material-selector> <option=cotton> <option=silk> <option=wool> <option=polyester> <option=linen> <option=leather> </material-selector> <material-properties> <slider=glossiness> <slider=roughness> <slider=weight> </material-properties> <simulation-preview> <text=drape-behavior> <text=texture-appearance> </simulation-preview> <button=apply-material> </MaterialSimView>	As a designer, I want to visualize how different fabrics behave on garments.	Simulate cotton, silk, wool, polyester, linen, leather properties. Show fabric drape and movement. Adjust glossiness, roughness, weight sliders. Real-time preview. Material-specific physics. Educational tooltips about fabric properties.	design

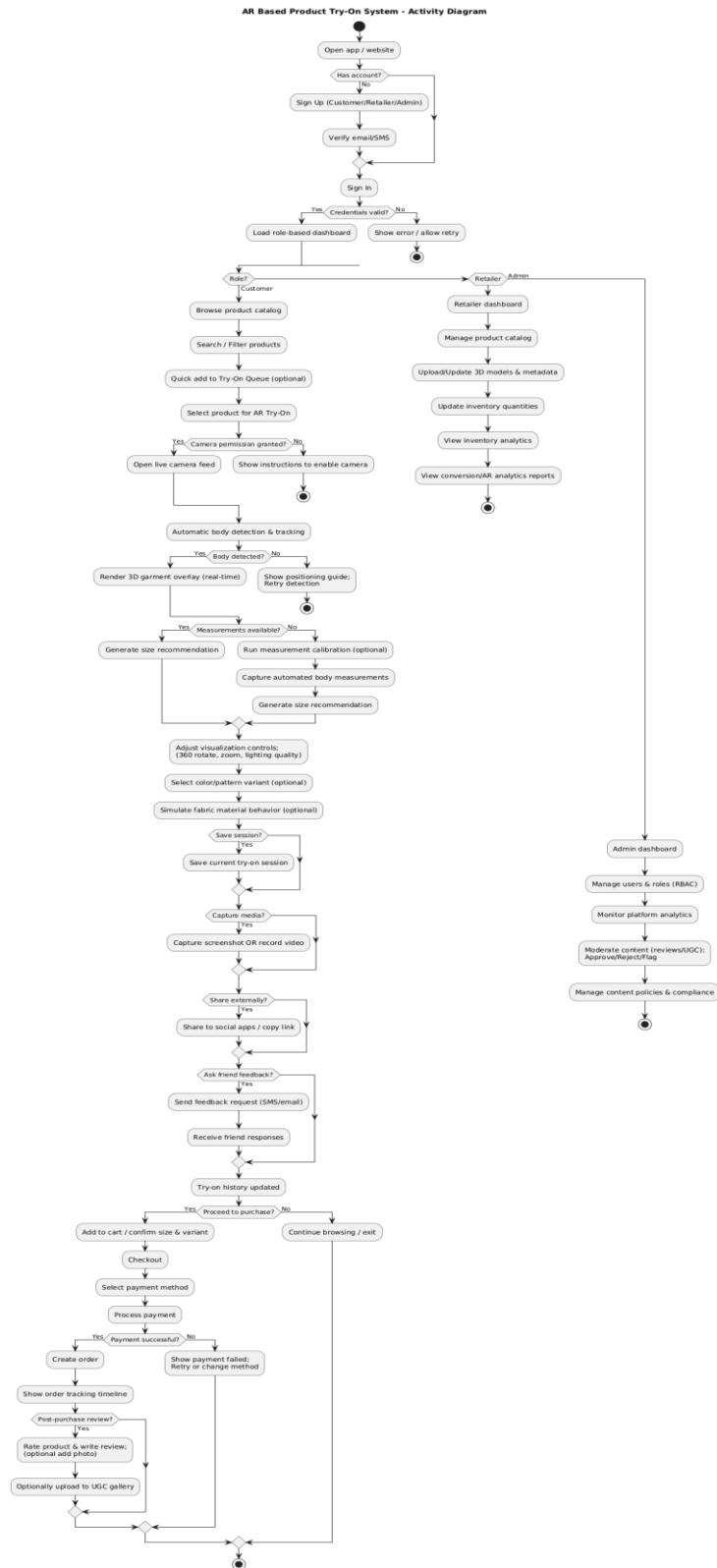
Feature: 10	Inventory & Stock Management					
phase 2	28	Real-Time Inventory Status Display [Customer] [Retailer]	<InventoryStatusView> <product-image /> <stock-indicator> <badge=in-stock> <badge=low-stock> <badge=out-of-stock> </stock-indicator> <size-availability> <table=size-stock-matrix> </size-availability> <restock-notification> <checkbox=notify-when-available> <text=estimated-restock-date> </restock-notification> <alternative-suggestions> <card=similar-product-1> </alternative-suggestions> </InventoryStatusView>	As a customer, I want to know product stock status before trying on.	Display stock badges (In Stock/Low Stock/Out of Stock). Show per-size availability in matrix. Real-time updates. Notify when out-of-stock items are restocked. Estimated restock dates. Suggest similar in-stock alternatives.	inventory
phase 2	29	Retailer Inventory Analytics Dashboard [Retailer] [Admin]	<InventoryAnalyticsDashboard> <overview-cards> <card=total-inventory> <card=low-stock-items> <card=fast-moving> <card=slow-moving> </overview-cards> <charts> <chart=stock-levels-trend> <chart=category-distribution> <chart=size-distribution> </charts> <inventory-alerts> <alert=reorder-required> <alert=overstocked> </inventory-alerts> <bulk-update> <button=import-csv> <button=export-report> </bulk-update> </InventoryAnalyticsDashboard>	As a retailer, I want comprehensive inventory insights and analytics.	Display total inventory, low-stock count, fast/slow-moving items. Show trends over time. Category and size distribution charts. Stock level alerts. Bulk update via CSV import. Export reports. Reorder recommendations.	analytics
Feature: 11	Payment Integration & Order Management					
phase 2	30	Secure Payment Gateway Integration [Customer] [Retailer]	<PaymentView> <order-summary /> <payment-methods> <option=credit-card> <option=debit-card> <option=mobile-wallet> <option=paypal> <option=buy-now-pay-later> </payment-methods> <card-details-form> <input=cardholder-name> <input=card-number> <input=expiry> <input=cvv> </card-details-form> <billing-address> <input=address> <input=city> <input=postal-code> </billing-address> <security-badge /> <button=process-payment> </PaymentView>	As a customer, I want a secure way to pay for products.	Support credit/debit cards, mobile wallets, PayPal, BNPL. PCI-DSS compliant encryption. Secure SSL/TLS connection. Fraud detection. Display security badges. Save payment methods securely. Process within 5 seconds.	payments
phase 2	31	Order Tracking & Fulfillment [Customer]	<OrderTrackingView> <order-header> <text=order-id> <text=order-date> <text=total-amount> </order-header> <order-items> <card=item-1> <image=product-image> <text=product-name> <text=quantity> <text=price> </card> </order-items> <tracking-timeline> <step=order-confirmed> <step=processing> <step=shipped> <step=delivered> </tracking-timeline> <tracking-map> <location-marker /> </tracking-map> <estimated-delivery> <text=delivery-date> </estimated-delivery> <button=contact-support> </OrderTrackingView>	As a customer, I want to track my orders in real-time.	Display order timeline (confirmed, processing, shipped, delivered). Show estimated delivery dates. Real-time location tracking. Delivery notifications. Provide tracking number. Contact support button. Return request option.	

Feature: 12		AI-Powered Recommendations & Analytics				
phase 2	32	Personalized Product Recommendations [Customer]	<RecommendationsView> <recommendation-sections> <section=based-on-history> <card=product-1> </section> <section=trending-now> <card=product-2> </section> <section=similar-style> <card=product-3> </section> <section=best-sellers> <card=product-4> </section> </recommendation-sections> <save-recommendation> <button=add-to-wishlist> <button=try-on-now> </save-recommendation> </RecommendationsView>	As a customer, I want personalized product recommendations.	ML-based recommendations from browsing history. Show trending and similar style items. Highlight best sellers. Update recommendations in real-time. Track recommendation click-through rates. A/B test recommendation engine.	ml
phase 2	33	Conversion & Analytics Dashboard [Retailer] [Admin]	<AnalyticsDashboard> <kpi-cards> <card=total-conversions> <card=conversion-rate> <card=average-order-value> <card=return-rate> </kpi-cards> <charts> <chart=daily-revenue> <chart=product-performance> <chart=user-funnel> <chart=ar-vs-non-ar-conversions> </charts> <user-behavior> <chart=page-heatmap> <chart=time-on-feature> </user-behavior> <export-options> <button=export-pdf> <button=export-csv> </export-options> </AnalyticsDashboard>	As a retailer, I want comprehensive analytics on conversions and performance.	Display KPIs (conversions, rate, AOV, returns). Show revenue trends and product performance. User funnel analysis. AR vs non-AR conversion comparison. Page heatmaps. Time-on-feature metrics. Export as PDF/CSV. Real-time dashboards.	analytics
Feature: 13		Loyalty & Rewards Program				
phase 2	34	Points-Based Loyalty System [Customer] [Retailer]	<LoyaltyView> <user-points> <text=current-points> <text=tier-status> <progress-bar=points-to-next-tier> </user-points> <points-breakdown> <text=points-from-purchases> <text=points-from-referrals> <text=points-from-reviews> </points-breakdown> <rewards-catalog> <card=reward-1> <text=reward-name> <text=points-required> <button=redeem> </card> </rewards-catalog> <tier-benefits> <card=bronze-tier> <card=silver-tier> <card=gold-tier> </tier-benefits> </LoyaltyView>	As a customer, I want to earn and redeem loyalty points.	Earn points on purchases (1 point per \$ spent), referrals, reviews. Display current balance and tier status. Show points breakdown by source. Tiered system (Bronze/Silver/Gold) with escalating benefits. Redeem for discounts/products. Expiration policy clear.	loyalty
phase 2	35	Referral Rewards Program [Customer]	<ReferralView> <referral-code> <text=your-code> <button=copy-code> <button=share-code> </referral-code> <referral-status> <text=successful-referrals> <text=earnings> </referral-status> <share-methods> <button=share-sms> <button=share-email> <button=share-social> </share-methods> <referral-rewards> <card=reward-1> <text=what-you-earn> <text=what-friend-earns> </card> </referral-rewards> </ReferralView>	As a customer, I want to earn rewards by referring friends.	Generate unique referral code for each user. Both referrer and referee get rewards. Track successful referrals. Share via SMS, email, social media. Display earnings from referrals. No referral limits.	social

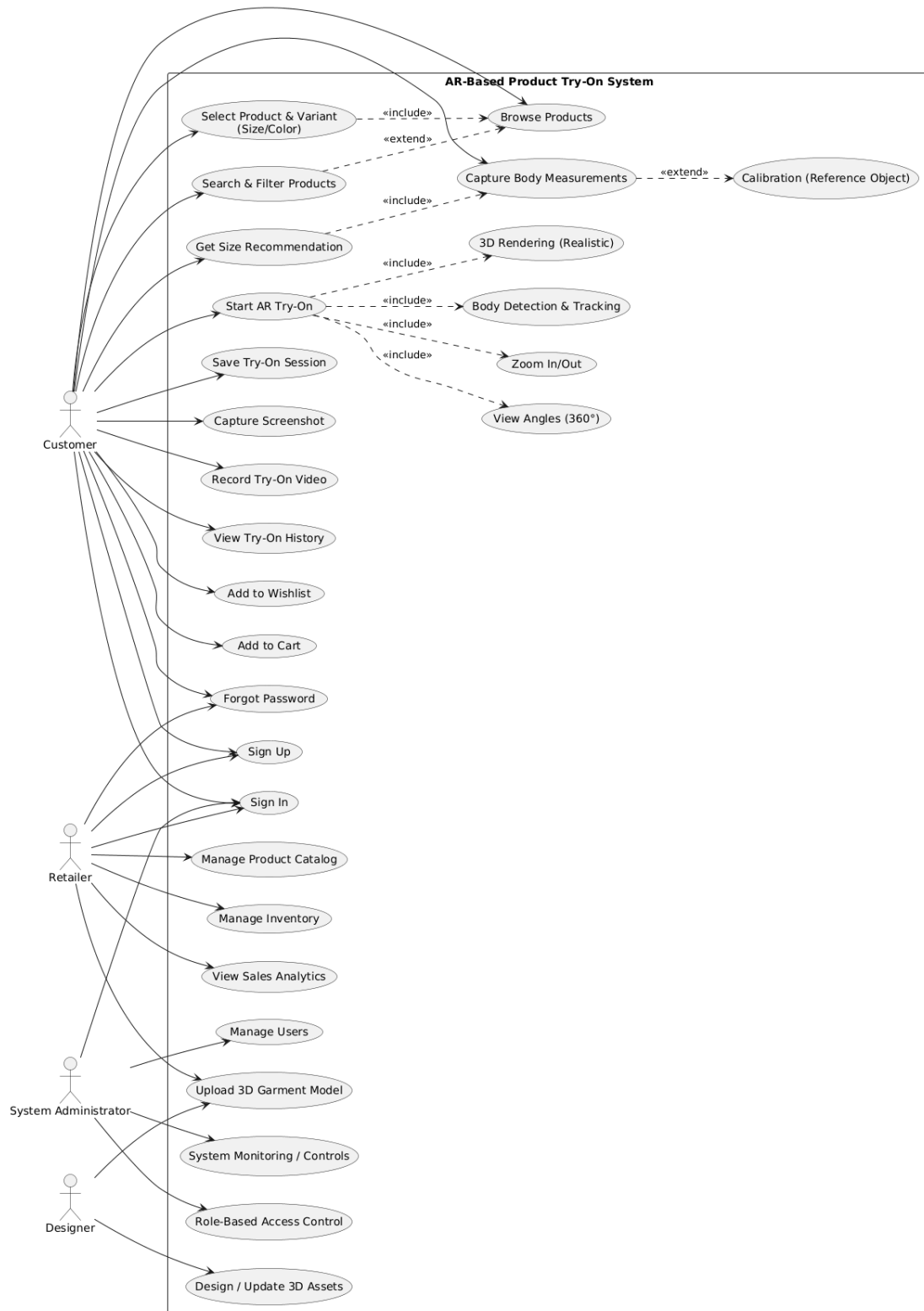
Feature: 14	Reviews & User-Generated Content					
phase 2	36	Product Reviews and Ratings [Customer]	<ReviewsView> <rating-summary> <text=average-rating> <stars=5-stars> <chart=rating-distribution> </rating-summary> <review-form> <rating-input=1-to-5-stars> <title-input> <text-area=review-text> <button=add-photo> <checkbox=recommend> <button=submit-review> </review-form> <reviews-list> <card=review-1> <text=reviewer-name> <stars=rating> <text=review-title> <text=review-text> <text=helpful-count> <button=mark-helpful> </card> </reviews-list> <filter-sort> <dropdown=sort-by> <!-- Helpful, Recent, Rating --> <dropdown=filter-by> </filter-sort> </ReviewsView>	As a customer, I want to read and write product reviews.	Display average rating and distribution chart. 1-5 star rating system. Title and text review with photos. Verified purchase badge. Mark reviews as helpful. Sort by helpful/recent/rating. Reply to reviews. Moderate spam/inappropriate content.	ugc
phase 2	37	User-Generated Try-On Photo Gallery [Customer]	<UGCGalleryView> <gallery-grid> <photo-card=user-photo-1> <image /> <text=user-name> <text=product-name> <rating=likes> <button=like> <button=view-product> </photo-card> </gallery-grid> <filter-options> <dropdown=sort-by> <!-- Trending, Latest --> <dropdown=filter-by-product> <dropdown=filter-by-size> </filter-options> <upload-photo> <button=upload-from-camera> <button=upload-from-gallery> </upload-photo> <moderation-status> <text=pending-approval> <text=approved> </moderation-status>	As a customer, I want to see how real users look in products.	Community gallery of user try-on photos. Display user name and product. Like/comment functionality. Sort by trending/latest. Filter by product/size. Upload from camera/gallery. Approve before display. Incentivize with points.	ugc
Feature: 15	Administration & Content Management					
phase 2	38	3D Model Upload & Management [Designer] [Retailer] [Admin]	<ModelUploadView> <upload-section> <file-input=3d-model-file> <input=product-sku> <input=product-name> <dropdown=category> <textarea=description> </upload-section> <preview> <3d-viewer /> <text=file-size> <text=supported-formats> </preview> <validation> <status=model-quality-check> <status=compatibility-check> </validation> <versioning> <text=current-version> <button=upload-new-version> </versioning> <button=publish> </ModelUploadView>	As a designer, I want to upload and manage 3D garment models.	Support FBX, OBJ, GLTF formats. Validate model quality and compatibility. Display preview in 3D viewer. Track versions with rollback option. Assign SKU and metadata. Publish/unpublish controls. Usage analytics.	admin
phase 2	39	Content Moderation & Compliance [Admin]	<ModerationDashboard> <queue-section> <card=user-review-1> <content-preview /> <flag-reason /> <button=approve> <button=reject> <button=flag-for-review> </card> </queue-section> <policy-management> <text=content-policies> <button=edit-policies> </policy-management> <banned-words> <table=word-filter-list> <button=add-word> </banned-words> <user-reports> <table=reported-content> <column=report-type> <column=action-taken> </user-reports> </ModerationDashboard>	As an admin, I want to moderate user content and ensure compliance.	Review queue for content moderation. Approve/reject user-generated content. Content policy enforcement. Banned words filter. Track reports and actions. User warnings/bans. Audit log for compliance. Auto-flag with ML.	admin

No.	Name	Non-Functional Requirements
NFR-1	Perform	The system shall respond to user interactions within 2 seconds under normal operating conditions.
NFR-2	Scalabi	The system shall support at least 1000 concurrent users without performance degradation.
NFR-3	Availab	The system shall maintain 99.5% uptime excluding scheduled maintenance.
NFR-4	Securit	The system shall encrypt all sensitive user data during transmission and storage.
NFR-5	Authen	The system shall require secure user authentication before accessing personalized features.
NFR-6	Privacy	The system shall comply with standard data privacy regulations and protect user personal information.
NFR-7	Usabili	The system shall be easy to use and understandable for first-time users without prior training.
NFR-8	User In	The system shall provide a clean, intuitive, and visually consistent user interface.
NFR-9	Compa	The system shall be compatible with modern Android and iOS devices.
NFR-10	Maintai	The system shall be designed to allow easy updates, bug fixes, and feature enhancements.
NFR-11	Reliabil	The system shall function correctly without failure during normal usage conditions.
NFR-12	Fault T	The system shall handle minor failures gracefully without crashing.
NFR-13	Loggin	The system shall log errors and system events for monitoring and debugging purposes.
NFR-14	Resour	The system shall minimize battery consumption and memory usage on mobile devices.
NFR-15	AR Acc	The system shall provide accurate real-time AR rendering of products.
NFR-16	Respor	The system shall load AR models within an acceptable time limit.
NFR-17	Data C	The system shall ensure consistency of user data across all modules.
NFR-18	Portab	The system shall be deployable across different mobile platforms with minimal changes.
NFR-19	Input H	The system shall handle invalid or unexpected user inputs without errors.
NFR-20	Commu	The system shall ensure secure communication between the client application and server.

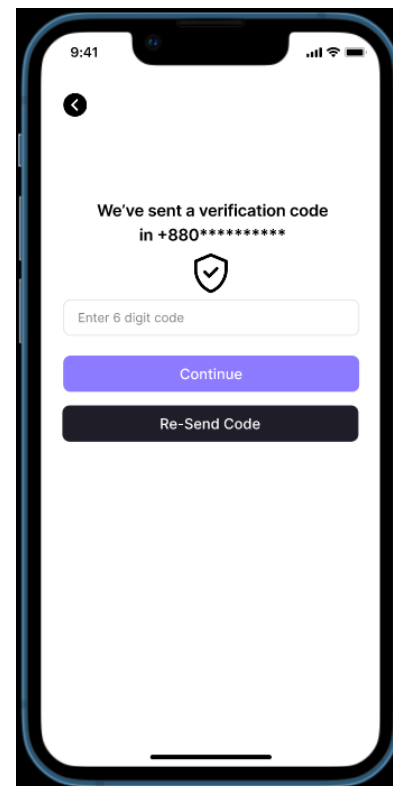
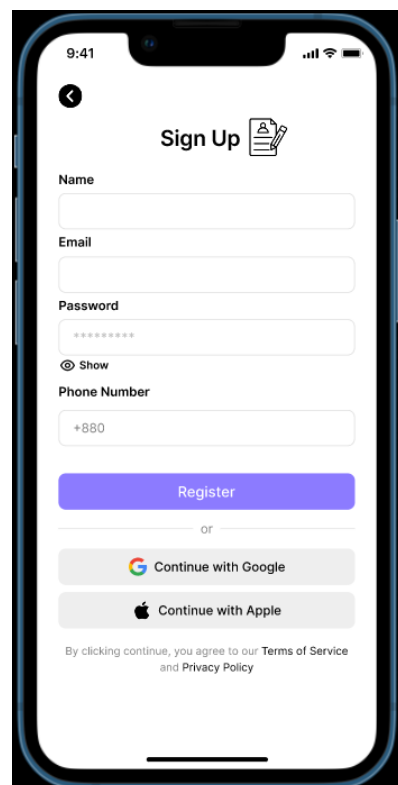
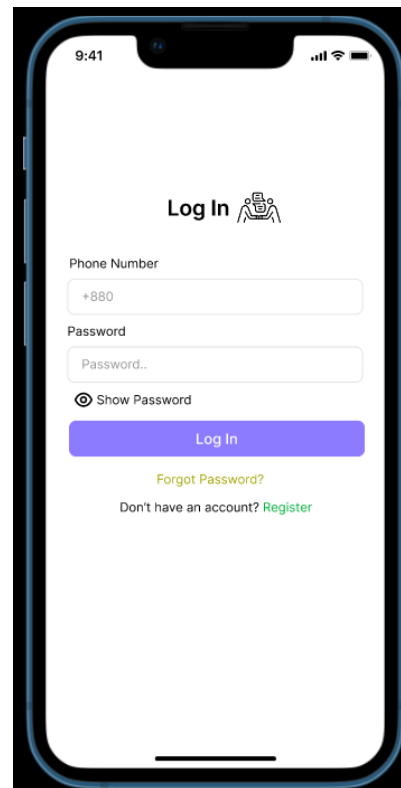
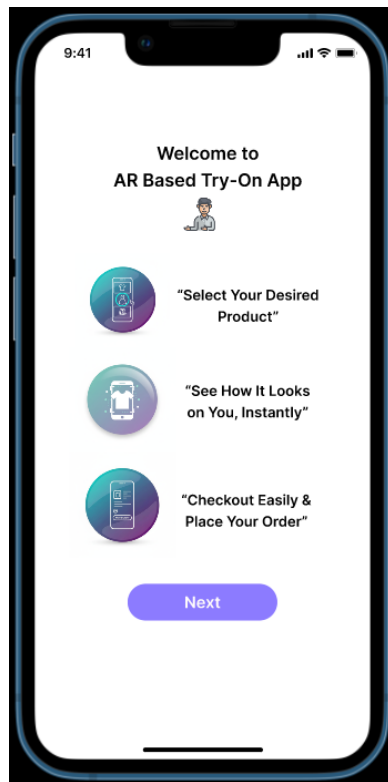
Activity Diagram

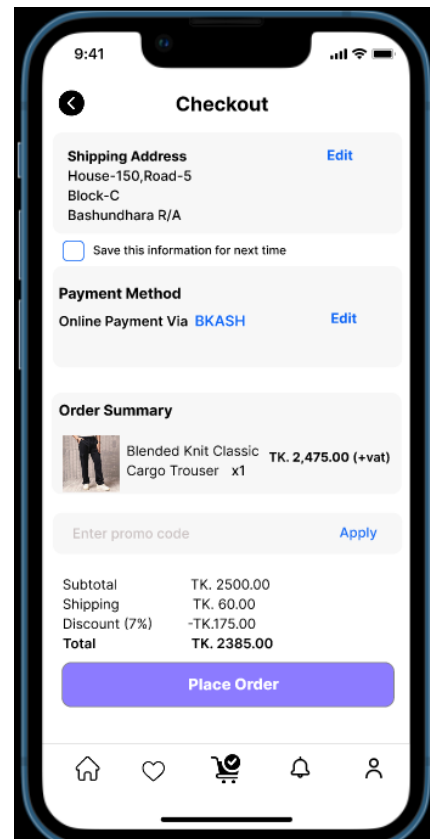
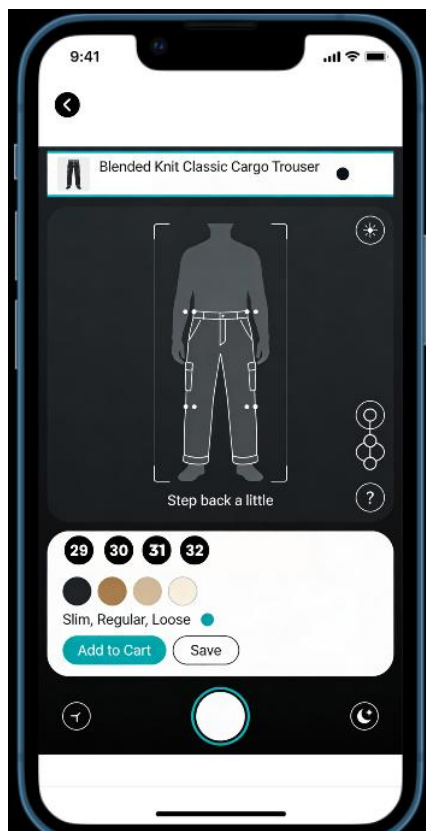
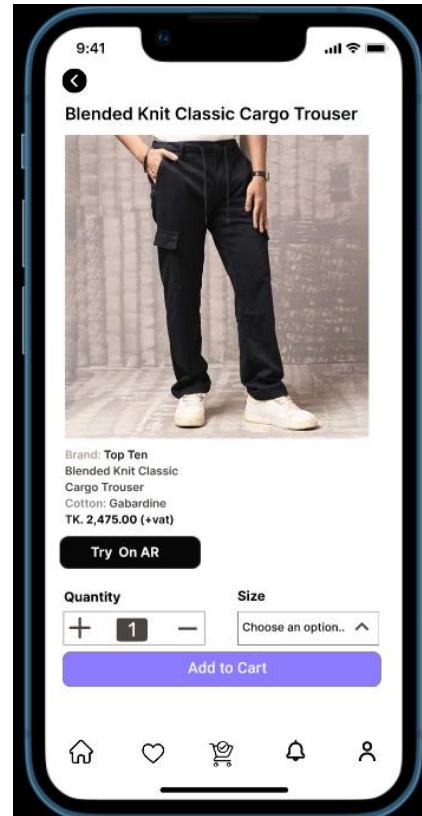
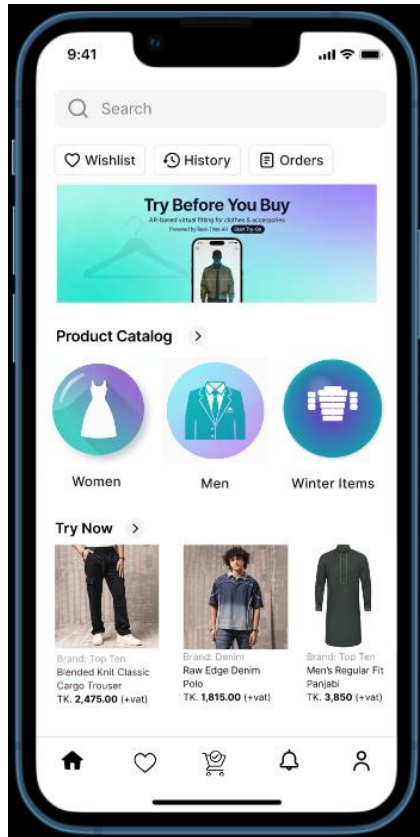


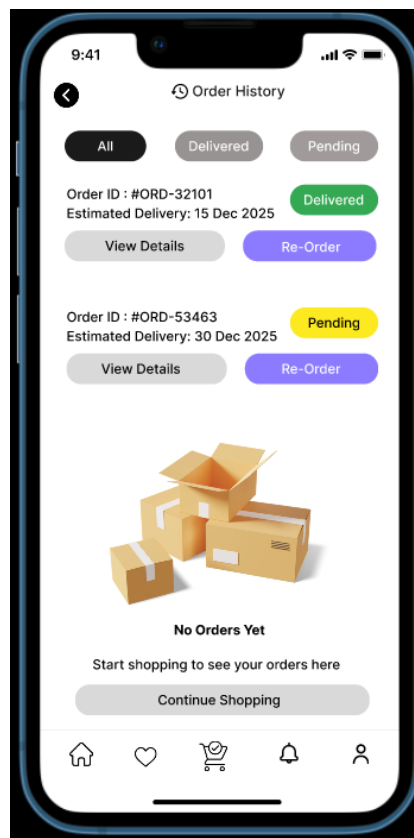
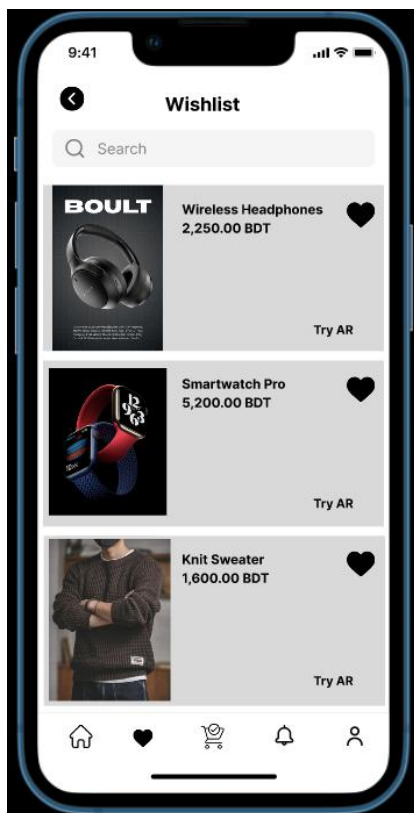
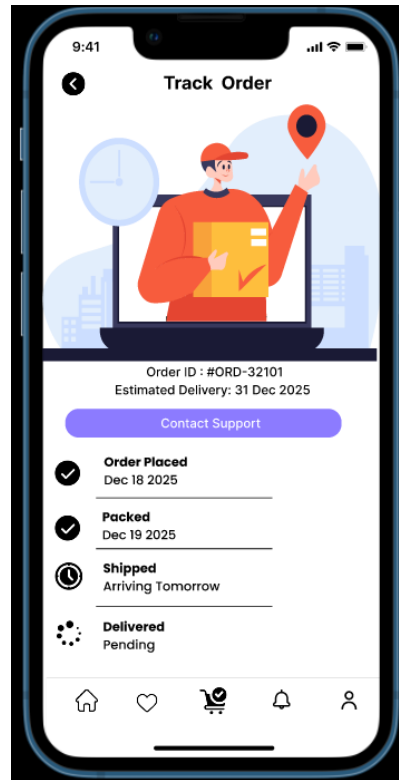
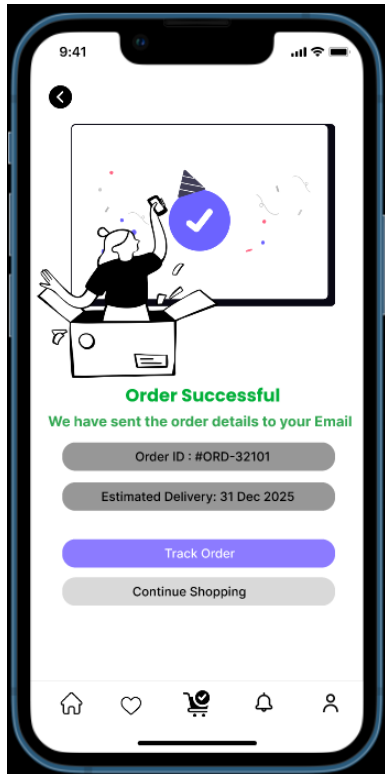
Use Case Diagram

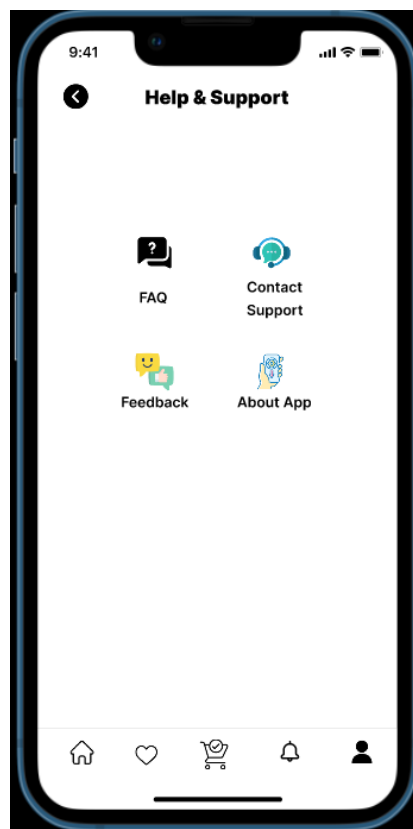
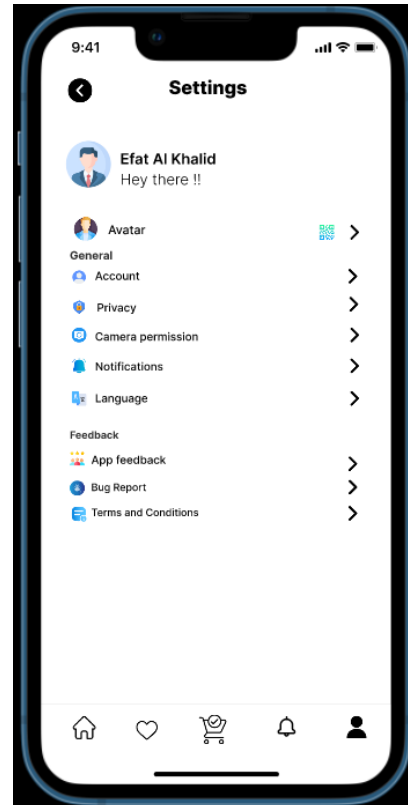
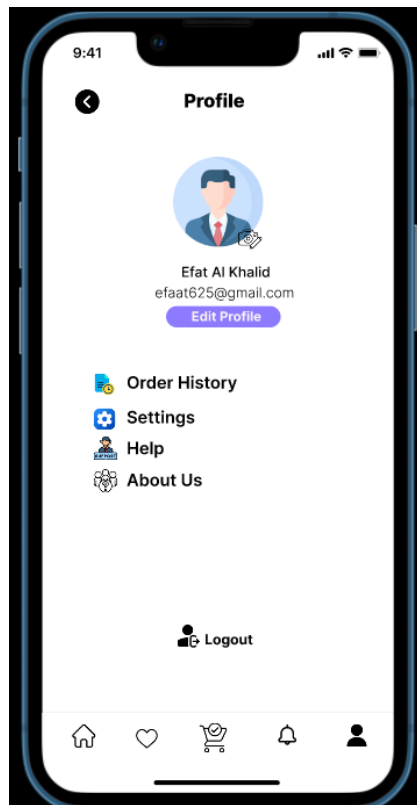


3.2 UI / Wireframe Design using Figma





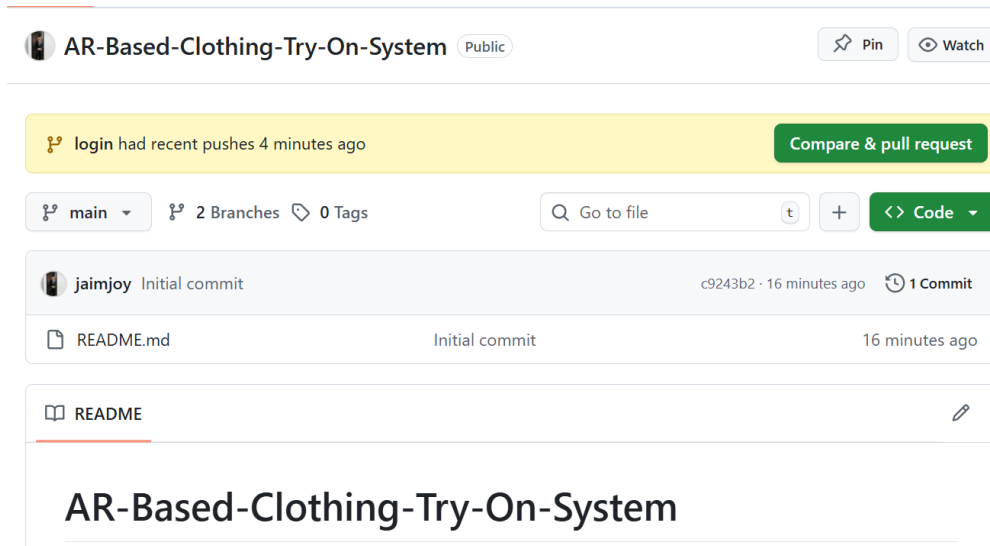




Figma Link: <https://www.figma.com/design/jQOvkyeR8CpZdxmwwMxNyB/AR-Based-Product-Try-On-System?node-id=0-1&t=xvQiiPK3wo7Wdhc-1>

4. GIT WORKFLOW

Repository Setup:



Member 1:

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Md. Hosne Jaim Joy>git clone https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System.git
Cloning into 'AR-Based-Clothing-Try-On-System'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Md. Hosne Jaim Joy>
C:\Users\Md. Hosne Jaim Joy>
C:\Users\Md. Hosne Jaim Joy>cd AR-Based-Clothing-Try-On-System
C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git checkout -b login
Switched to a new branch 'login'

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git branch
* login
  main

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git status
On branch login
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Activity Diagram.png

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git add .
C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git commit -m "Login feature will be added"
[login 724f215] Login feature will be added
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Activity Diagram.png
C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git push origin login
Enumerating objects: 4, done.
```

```

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git checkout login
Switched to branch 'login'

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git merge main
Already up to date.

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git checkout Signup
error: pathspec 'Signup' did not match any file(s) known to git

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git branch
* login
  main

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git pull origin main
From https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System
* branch      main      -> FETCH_HEAD
Already up to date.

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git merge login
Updating c9243b2..724f215
Fast-forward
 Activity Diagram.png | Bin 0 -> 91824 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Activity Diagram.png

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System.git
 c9243b2..724f215  main -> main

```



AR-Based-Clothing-Try-On-System Public



Pin



Watch

Signup ▾

3 Branches 0 Tags

Go to file

t

+

<> Code ▾

This branch is **2 commits ahead of** and **2 commits behind** `main`.

Contribute ▾



efat-khalid use case up

87fecf6 · 17 minutes ago

3 Commits



Class Diagram.jpeg

signup add

34 minutes ago



README.md

Initial commit

48 minutes ago



Use case diagram.jpeg

use case up

17 minutes ago

```

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git fetch -all
error: did you mean '--all' (with two dashes)?

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git fetch --all

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git branch -a
* Signup
  login
  main
remotes/origin/HEAD -> origin/main
remotes/origin/Signup
remotes/origin/login
remotes/origin/main

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git log --oneline --graph --all
* 8e989e1 (origin/main, origin/HEAD, main) Class Diagram
* 724f215 (origin/login, login) Login feature will be added
| * 87fecf6 (HEAD -> Signup, origin/Signup) use case up
| * 5ece712 signup add
|/
* c9243b2 Initial commit

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>git shortlog -s -n --all
    2 efat-khalid
    2 jaimjoy
    1 MD Hosne Jaim Joy

C:\Users\Md. Hosne Jaim Joy\AR-Based-Clothing-Try-On-System>

```

Member 2:

```

Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Efat>git clone https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System.git
Cloning into 'AR-Based-Clothing-Try-On-System'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Efat>cd AR-Based-Clothing-Try-On-System

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git checkout -b Signup
Switched to a new branch 'Signup'

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git brance
git: 'brance' is not a git command. See 'git --help'.

The most similar command is
    branch

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch
* Signup
  main

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git status
On branch Signup
nothing to commit, working tree clean

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git status
On branch Signup
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Class Diagram.jpeg

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git add .

```

```

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git push origin Signup
remote: Permission to jaimjoy/AR-Based-Clothing-Try-On-System.git denied to efat-khalid.
fatal: unable to access 'https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System.git/': The requested URL returned
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch
* Signup
  main
C:\Users\Efat\AR-Based-Clothing-Try-On-System>cd AR-Based-Clothing-Try-On-System
The system cannot find the path specified.
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch
* Signup
  main
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git checkout Signup
Already on 'Signup'
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git status
On branch Signup
nothing to commit, working tree clean
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git status
On branch Signup
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    Class Diagram.jpeg

no changes added to commit (use "git add" and/or "git commit -a")
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git status
On branch Signup
nothing to commit, working tree clean

```

```

Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 330.30 KiB | 11.80 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'Signup' on GitHub by visiting:
remote:   https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System/pull/new/Signup
remote:
To https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System.git
 * [new branch]      Signup -> Signup

C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch
* Signup
  main
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git fetch origin
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 6 (delta 1), reused 5 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 251.86 KiB | 877.00 KiB/s, done.
From https://github.com/jaimjoy/AR-Based-Clothing-Try-On-System
 * [new branch]      login    -> origin/login
 c9243b2..8e989e1    main     -> origin/main
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch -r
  origin/HEAD -> origin/main
  origin/Signup
  origin/login
  origin/main
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git checkout -b login origin/login
branch 'login' set up to track 'origin/login'.
Switched to a new branch 'login'
C:\Users\Efat\AR-Based-Clothing-Try-On-System>git branch
  Signup
* login
  main

```


5. SOFTWARE TESTING

Project Name: AR-Based Clothing Try-On System		Test Designed by: Afia Mushfika Shrabosty		
Test Case ID: TC_01		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Afia Mushfika Shrabosty		
Module Name: Registration (Sign Up)		Test Execution date: 03/01/2026		
Test Title: Sign up with existing email/phone				
Description: Verify sign up fails when the email/phone is already registered.				
Precondition: User is on Sign Up page				
Dependences: Existing account with same email/phone is present in database				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open sign up page. 2. Enter name, email/phone that is already registered. 3. Enter valid password (>=8 characters). 4. Enter same value in confirm password. 5. Select user type. 6. Accept terms and conditions. 7. Click “Sign Up”.	Name: Md. Hosne Jaim Joy Email: jaim81200@gmail.com Phone: 01719139734 Password: Abcd1234@ Confirm Password: Abcd1234@ User Type: Customer	System shows error message “Email/phone already in use” and does not create account.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Afia Mushfika Shrabosty		
Test Case ID: TC_02		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Afia Mushfika Shrabosty		
Module Name: Authentication (Sign In)		Test Execution date: 03/01/2026		
Test Title: Account locks after 3 failed sign-in attempts				
Description: Verify account is temporarily locked after maximum 3 incorrect password attempts.				
Precondition: Registered user exists; user is on sign in page.				
Dependencies: Auth service and user database available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open sign in page. 2. Enter registered email/phone. 3. Enter wrong password and click “Sign In”. 4. Repeat step 3 until total 3 failed attempts.	Email: jaim81200@gmail.com Password: Abcd1234@ Remember Me: Unchecked	System shows “Invalid password”; after 3 failed attempts, system temporarily locks the account.	Account not locked after 3 attempts (lock not triggered).	Fail

Project Name: AR-Based Clothing Try-On System		Test Designed by: Afia Mushfika Shrabosty		
Test Case ID: TC_03		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Afia Mushfika Shrabosty		
Module Name: AR Try-On Core (Camera Permission)		Test Execution date: 03/01/2026		
Test Title: Show error when camera permission is denied				
Description: Verify system shows a clear error with instructions when camera permission is denied.				
Precondition: User is attempting to start AR Try-On.				
Dependences: Device permission prompt available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open any product details page. 2. Tap “Try On AR”. 3. Deny camera permission.	Camera Permission: Denied	System displays an error message guiding user to enable camera permission from device settings.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Shahadout Md. Efat Al Khalid		
Test Case ID: TC_04		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): High		Test Executed by: Shahadout Md. Efat Al Khalid		
Module Name: AR Try-On Core (Overlay Performance)		Test Execution date: 03/01/2026		
Test Title: AR overlay appears within 2 seconds				
Description: Verify camera feed starts immediately and AR garment overlay appears within 2 seconds of product selection.				
Precondition: Camera permission granted.				
Dependences: Product has AR model available; AR service running.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open product that supports AR Try-On. 2. Tap “Try On AR”. 3. Select the garment if required. 4. Observe overlay load time.	Product: Any AR-enabled item Network: Stable	Camera feed displays immediately; AR overlay appears within 2 seconds; overlay updates smoothly (min 30 fps) and stays aligned.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Shahadout Md. Efat Al Khalid		
Test Case ID: TC_05		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Shahadout Md. Efat Al Khalid		
Module Name: Body Measurement Detection		Test Execution date: 03/01/2026		
Test Title: Measurements captured within 10 seconds after guided poses				
Description: Verify body measurements are calculated within 10 seconds and saved to profile.				
Precondition: User is on measurement screen with camera active.				
Dependences: Body detection working; good lighting.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Body Measurement feature. 2. Follow step-by-step pose guide. 3. Complete guided poses. 4. Wait for measurement calculation. 5. Tap “Save Measurements”.	Lighting: Normal Pose: As instructed	System calculates height, chest, waist, hips, shoulder width within 10 seconds; accuracy target 95%; values saved to profile and remain editable.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Shahadout Md. Efat Al Khalid		
Test Case ID: TC_06		Test Designed date: 03/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Shahadout Md. Efat Al Khalid		
Module Name: Screenshot Capture		Test Execution date: 03/01/2026		
Test Title: Capture screenshot within 0.5 seconds with preview options				
Description: Verify screenshot capture timing, preview, and save/share/delete options.				
Precondition: Active AR try-on session running.				
Dependencies: Storage permission granted; storage available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start an AR Try-On session. 2. Tap Capture (camera) button. 3. Observe preview thumbnail. 4. Select Save/Share/Delete options.	Format: JPEG/PNG Storage: Available	Screenshot captured within 0.5 seconds; saved as JPEG/PNG; thumbnail preview appears immediately with save/share/delete; stored in app gallery with metadata (product name, date, time).	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Md. Hosne Jaim Joy		
Test Case ID: TC_07		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Md. Hosne Jaim Joy		
Module Name: Product Catalog (Filter & Search)		Test Execution date: 04/01/2026		
Test Title: Search returns results within 2 seconds				
Description: Verify product search returns relevant results within 2 seconds.				
Precondition: User is logged in and on Product Catalog screen.				
Dependences: Product database populated; network available				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Product Catalog. 2. Tap search bar. 3. Enter a valid query. 4. Tap Search/Submit. 5. Observe response time and results.	Query: “jacket”; Network: Stable	Results appear within 2 seconds; results are relevant to query; if none, “No results found” message shown with suggestions.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Md. Hosne Jaim Joy		
Test Case ID: TC_08		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Md. Hosne Jaim Joy		
Module Name: Product Visualization Controls (Viewing Angles)		Test Execution date: 04/01/2026		
Test Title: Front/Side/Back view switches within 0.5 seconds				
Description: Verify quick-access buttons switch viewing angles (front/side/back) within 0.5 seconds and remain aligned				
Precondition: Active AR try-on session running with garment visible				
Dependences: AR tracking stable; supported device performance.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start AR Try-On. 2. Tap “Front view”. 3. Tap “Side view”. 4. Tap “Back view”. 5. Observe switching speed and alignment.	Device: Mid-range; Network: Any	Each view switches within 0.5 seconds; rotation is smooth; garment stays aligned to body during switching.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Md. Hosne Jaim Joy		
Test Case ID: TC_09		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Md. Hosne Jaim Joy		
Module Name: Size Recommendation (Fit Analysis)		Test Execution date: 04/01/2026		
Test Title: Size recommendation appears within 1 second using saved measurements				
Description: Verify the system recommends a size within 1 second when measurements exist, including confidence and explanation				
Precondition: User has saved body measurements in profile; product has size chart.				
Dependencies: Measurement service available; retailer size chart available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Customer. 2. Open a product with multiple sizes. 3. Navigate to size recommendation area. 4. Observe recommendation content and timing.	Measurements: Saved; Product: Has size chart	Recommendation shown within 1 second; includes recommended size, confidence % and brief reason; shows alternative sizes if between sizes.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Iftekhar Ahmed Antu		
Test Case ID: TC_10		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Iftekhar Ahmed Antu		
Module Name: Try-On Session Management (Save Session)		Test Execution date: 04/01/2026		
Test Title: Save try-on session completes within 2 seconds and is retrievable				
Description: Verify saving a try-on session completes within 2 seconds, stores selections (size/color) and can be accessed later.				
Precondition: Active try-on session with a product selected (and size/color chosen if available).				
Dependences: User profile storage available; network stable (if cloud sync is used).				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Start AR Try-On for any product. 2. Select size and color (if applicable). 3. Tap “Save session”. 4. Provide optional session name. 5. Open “Saved Try-Ons” from profile. 6. Verify saved entry and reload it..	Session name: “Winter look”; Network: Stable	Save completes within 2 seconds; confirmation message shown; saved session includes product, size, color, settings, timestamp; saved session is visible and reload restores previous settings.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Iftekhar Ahmed Antu		
Test Case ID: TC_11		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Iftekhar Ahmed Antu		
Module Name: AR Try-On Session Management (Try-On History)		Test Execution date: 04/01/2026		
Test Title: Try-on history stores sessions and reloads a previous try-on				
Description: Verify the system stores try-on sessions with timestamp/product details and can reload a selected history item with previous settings restored				
Precondition: User is logged in; at least one try-on session completed.				
Dependences: History storage available; product still accessible (or marked unavailable if not).				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Perform a try-on for Product A. 2. Exit try-on. 3. Open Try-On History screen. 4. Select the saved history item. 5. Verify session reload.	Product A: Any AR-enabled item	History entry appears with product details and timestamp; selecting entry reloads the try-on session with previous settings restored; list order is reverse chronological.	As expected	Pass

Project Name: AR-Based Clothing Try-On System		Test Designed by: Iftekhar Ahmed Antu		
Test Case ID: TC_12		Test Designed date: 04/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed by: Iftekhar Ahmed Antu		
Module Name: Payment Integration (Checkout)		Test Execution date: 04/01/2026		
Test Title: Payment processing completes within 5 seconds				
Description: Verify payment flow supports payment methods and processes payment within 5 seconds under normal conditions.				
Precondition: User has items in cart and is on Payment screen.				
Dependencies: Payment gateway available; secure connection enabled (SSL/TLS).				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Add product to cart. 2. Proceed to checkout. 3. Select a payment method. 4. Enter valid payment details. 5. Tap “Process Payment”. 6. Measure completion time	Method: Card/Mobile wallet; Network: Stable	Payment is processed within 5 seconds; success message shown; order is created; user is redirected to confirmation/order details.	As expected	Pass

6. CONCLUSION

This AR Virtual Try-On product provides an immersive, practical way to preview how apparel and accessories look on a person before purchasing, using a live camera view and real-time overlays to reduce uncertainty in online shopping. Instead of limiting the experience to trousers, the same workflow can support a broad catalog—tops, bottoms, dresses, outerwear, shoes, and even accessories—so users can evaluate style, proportions, and overall appearance across many product types in one consistent interface. Clear on-screen guidance (alignment and tracking cues) and fast controls for size, color, and fit encourage exploration while keeping the try-on process simple and engaging. By helping customers make more informed decisions, virtual try-on can increase purchase confidence and reduce the likelihood of returns, which benefits both shoppers and retailers. The add-to-cart flow built into the try-on screen creates a smooth path from visualization to checkout, supporting stronger conversion and a modern shopping experience. Overall, this solution is a scalable foundation for interactive fashion retail, with room to extend into personalized recommendations and more realistic garment behavior as tracking and sizing models improve.